

Question	Answer	Mark	Mark scheme	Additional guidance
16 (a)	4.9	P1	for beginning to work with scale factors eg $\left(\frac{h}{12}\right)^3$ or $\left(\frac{12}{h}\right)^3$ or $\frac{90}{1350} \left(=\frac{1}{15}\right)$ or $\frac{1350}{90} (=15)$ or $90 : 1350$ oe P1 for equating scale factors eg $\left(\frac{h}{12}\right)^3 = \frac{90}{1350}$ or $\left(\frac{12}{h}\right)^3 = "15"$ or for finding the linear scale factor, eg $\sqrt[3]{\frac{90}{1350}}$ or $\sqrt[3]{"15"} (=2.46(6\dots))$ P1 for a complete method eg $\sqrt[3]{12^3 \times \frac{90}{1350}}$ or $12 \times \sqrt[3]{\frac{90}{1350}}$ or $\frac{12}{\left(\sqrt[3]{"15"}\right)}$ A1 for answers in the range 4.8 to 4.9	
(b)	$(15)^{\frac{2}{3}}$	B1	for $15^{\frac{2}{3}}$ or $\left(\sqrt[3]{15}\right)^2$ or $\sqrt[3]{15^2}$ or $225^{\frac{1}{3}}$	If given in an acceptable form, then given as a decimal, isw and award